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ECEN3019 INTRODUCTION TO RADIO-FREQUENCY CIRCUITS AND SYSTEMS

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Simulation Report – LNA Design

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Abstract— To further discuss how the low-noise amplifier (LNA) works, a LNA that meets the specifications to have great input/output matching and linearity, low noise figure (NF), and reasonable power gain with gain switching applied is designed in Cadence for this project. Both the theoretical computation process and the simulation results of the device parameters are shown in the report as follows. These parameters require careful optimization to achieve a balanced configuration that ensures optimal amplifier performance.

Index Terms—Low-Noise Amplifier (LNA), Center frequency, Input/Output Matching, Power gain, Noise Figure (NF), 1dB Compression Point (P1dB), Third Order Intercept Point (IIP3), Power Consumption.

I. INTRODUCTION

THE LNA, which can amplify weak signals without significantly degrading the signal-to-noise ratio (SNR), plays a crucial role in the radio-frequency systems. By using LNAs in the front-end, with a low NF achieved, the sensitivity of the entire receiving system can be improved.

For the LNA design in this project, it's preferred to get higher gain, greater linearity, and lower NF within the specific power consumption requirement. The parameters' specifications are shown in Table 1. In addition, the parameters should also satisfy $Q \leq 15$ and $L \geq 200\text{pH}$.

TABLE I
SPECIFICATIONS FOR THE LNA

High-gain Mode		
Parameters	Specifications	Comments
Frequency Range	4.9 - 5.1 GHz	
Input/Output Matching (S11 and S22)	$\leq -10\text{dB}$	
Power Gain (S21)	$\geq 20\text{dB}$	
NF	$\leq 1.5\text{dB}$	Input power at -60dBm
P1dB	$\geq -36\text{dBm}$	Input frequency at 5GHz
IIP3	$\geq -26\text{dBm}$	Two-tone frequencies at 4.98GHz and 5.02GHz
Supply Voltage	1V	
Power Consumption	$\leq 3.8\text{mW}$	
Low-gain Mode		
Parameters	Specifications	Comments
Frequency Range	4.9 - 5.1 GHz	
Input/Output Matching (S11 and S22)	$\leq -10\text{dB}$	
Power Gain (S21)	$\leq 10\text{dB}$	
NF	$\leq 4\text{dB}$	Input power at -60dBm
P1dB	$\geq -26\text{dBm}$	Input frequency at 5GHz
IIP3	$\geq -16\text{dBm}$	Two-tone frequencies at 4.98GHz and 5.02GHz
Supply Voltage	1V	
Power Consumption	$\leq 4.2\text{mW}$	

Frequency range is determined to amplify the signal within 4.9 – 5.1 dB, where 5dB is set as the central frequency ω_0 . Input/Output matching is necessary for the LNA to achieve maximum signal power transfer and the minimum NF. However, the optimal points for these two requirements may differ, that's where we need to make the trade-offs. S11 and S22 are the reflection loss for the input and output accordingly. Poor input/output matching causes signal reflection, reducing effective received power. P1dB and IIP3 both indicates the linearity performance of the LNA. P1dB symbolizes the onset of gain compression, where the output power no longer increases linearly with input power, and IIP3 relates to the intermodulation distortion (IMD) performance, particularly for two-tone scenarios.

Specifications for both high-gain and low-gain mode should be achieved in this project. The ability to switch between high-gain mode and low-gain Mode helps to optimize the linearity and noise performance under different input signal conditions. As shown in Figure 1, high-gain mode maximizes sensitivity for weak signals at the cost of linearity, while low-gain mode preserves linearity for strong signals but sacrifices sensitivity as more noise is introduced.

With a 1V supply voltage and power consumption within 3.8mW (for the high-gain mode) or 4.2mW (for the low-gain mode), the device parameters and the circuit of the designed LNA are shown in the following part.

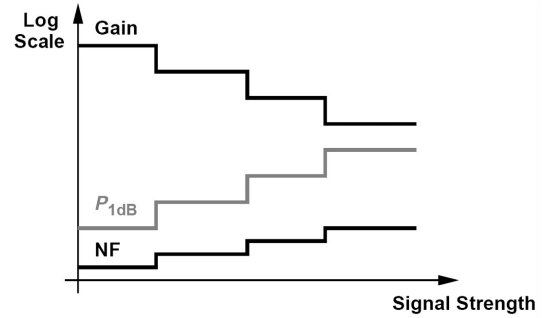


Fig. 1. The role of gain switching.

II. CIRCUIT IMPLEMENTATION

For the LNA designed in this project, the detailed device parameters are shown in Table 2, and the structure of the final circuit is shown in Figures 2 and 3 below.

TABLE II
DEVICE PARAMETERS TABLE

MOSFET Parameters				
MOSFETs	Length (m)	Total Width(m)	Finger width (m)	Fingers
NM0	60n	32u	1u	32
NM1	60n	1u	1u	1
NM2	60n	1u*W=36u	1u	W=36
NM3	60n	10u	1u	10
NM5	60n	32u	1u	32

Parameters of Variables (as shown in Figure 2)			
Variables	Parameters		
VDD	1 V (DC Voltage)	R0	100K Ω
VSS	0V	R1	80 Ω
VB	0V (high-gain mode) / 1V (low-gain mode)	$Q_{Lg}/Q_{Ld}/Q_{Ls}$	15
frf	5G	Lg	5.5n H
prf	-60	Ld	10.5n H
C1/C2/C3	10n F	Ls	900p H
		Cs	142f F
		Cd	56f F
		Ib	58u A (DC current)

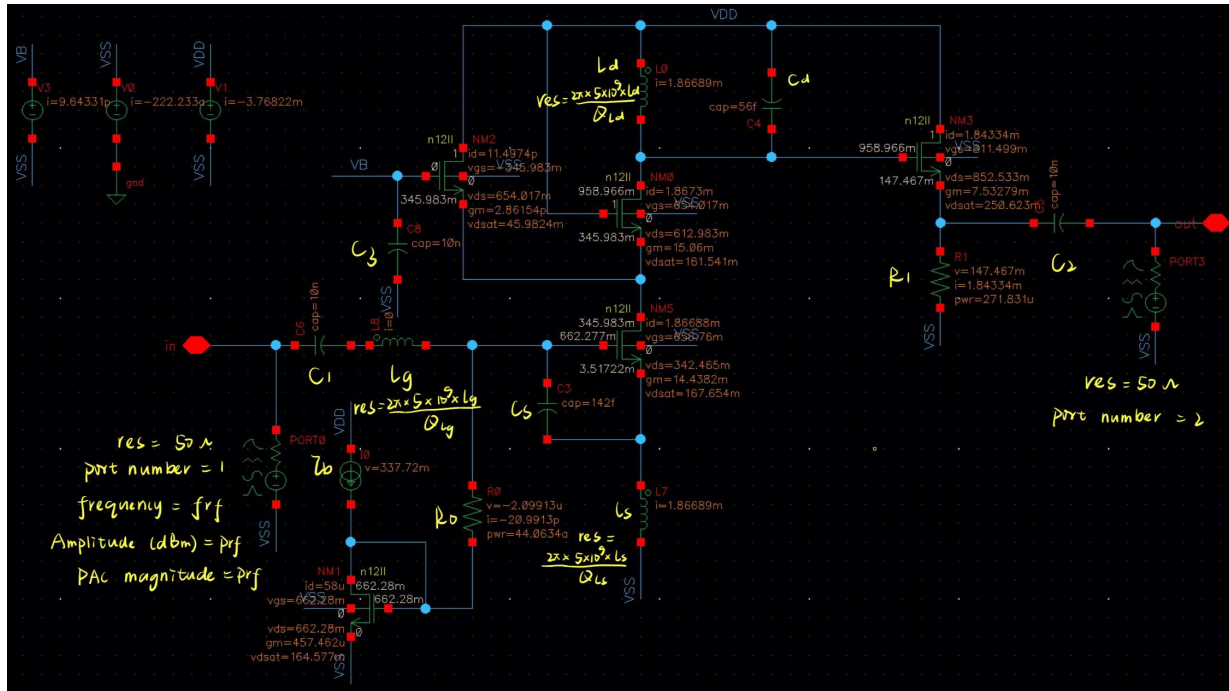


Fig. 2. Circuit Diagram for the LNA in high-gain mode.

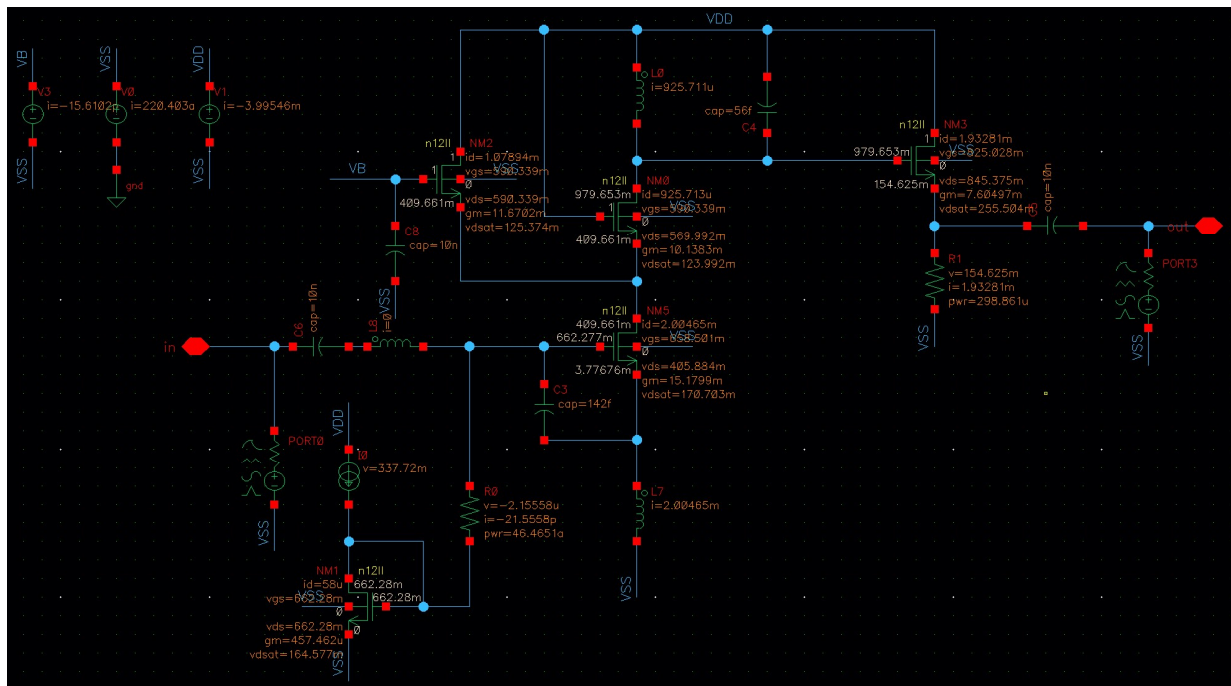


Fig. 3. Circuit Diagram for the LNA in low-gain mode.

III. DESIGN METHODOLOGY

Ports 0 and 3, which are also numbered as 1 and 2, are used as the input and output stages. Both stages have the resistance equal to 50Ω , which is the general standard in the field of radio frequency (RF). Especially for the input stage, the input power is set as $\text{prf} = -60\text{dBm}$ and the input frequency is set to $\text{frf} = 5\text{G}$ Hz. In addition to filtering out high-frequency noise and maintaining local voltage stability, C1, C2, and C3 are set to 10nF to be used as coupling capacitors. Furthermore, the lengths of all the MOSFETs in this RF circuit project are set to the smallest value of 60nm to achieve the fastest switching speed (though the short-channel effect is worse, it's not such a crucial point in the RF field).

The design procedure for this LNA can be separated into mainly 5 parts: the amplifier design, the center frequency adjustment, Input/Output matching, the gain switching design, and the final fine-tuning to achieve all the specifications.

To begin with, a Cascode amplifier (NM0 & NM5) connected with a Source Follower (NM3) is used as a two-stage amplifier in this LNA structure. The Cascode structure is widely used in LNAs, which combines the transconductance (g_m) of the common-source (CS) stage with the current buffering of the common-gate (CG) stage. It has advantages, including higher overall gain, improved linearity, low NF, and so on. The source follower in this circuit primarily isolates the output impedance from the effects of inductance L_d and capacitance C_d , ensuring it's determined by resistor R_1 and MOSFET NM3.

Based on the power consumption specifications, which indicate that the current flow through each branch should be maintained between 1 and 2 mA. To ensure gain, the current should not be set too low either. In this case, adjust the current mirror formed between NM1 and NM5 to provide a voltage around 662mV that makes NM5 saturated. Also, make sure other MOSFETs are saturated as well by varying the width, length, fingers, and other settings.

The bandwidth and center frequency are determined mainly by L_d and C_d , which should satisfy:

$$\omega_0 = 2\pi f_0 = \pi \times 10^{10} = \frac{1}{\sqrt{L_d \times C_d}} \text{ rad/s} \quad (1)$$

Q_{L_d} is the quality factor of the inductor L_d , which is related to the bandwidth of the LNA. With a higher Q factor, the overall gain is higher, while the bandwidth will be relatively narrow. This problem can be improved by connecting a resistor R_{SL} in series beside the inductor, where

$$R_{SL} = \frac{2\pi \times 5G \times L}{Q} \Omega \quad (2)$$

$$R_P = (Q_L^2 R_{SL}) || (Q_C^2 R_{CL}) \approx Q_L^2 R_{SL} \Omega \quad (3)$$

$$\text{Gain}(\omega_0) = g_m \times R_P \quad (4)$$

Where R_P is the equivalent resistor that parallels to the L_d and C_d , Q_C is assumed to be infinity and R_{CL} is the resistor connected in series with the capacitor C_d and not set in this

project. It's also quite interesting to notice that the performance of linearity will become worse if the value of L_d or C_d gets larger. In this case, let $Q_L = 15$ to get a larger gain and set $L_d = 10.5\text{nH}$ and $C_d = 56\text{fF}$. Therefore, the center frequency can be calculated:

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{\sqrt{L_d \times C_d}} = \frac{1}{\sqrt{10.5n \times 56f}} = 6.56\text{G Hz} \quad (5)$$

Which still has some differences compared to the simulation result where $f_0 = 5.32\text{GHz}$ (not 5G since other specifications should be satisfied). This is because the frequency may also be influenced by other values and shows the existence of errors in theoretical and practical simulations.

Moreover, the input and output matching is typically the most important consideration of the LNA design. According to the lecture note 9:

$$Z_{in(ideal)} = \frac{1}{sC_s} + s(L_g + L_s) + \frac{g_m L_s}{C_s} = 0s + 50 \Omega \quad (6)$$

$$s = 2\pi f_0 \quad (7)$$

Where the real part impedance = 50Ω is provided by L_s and C_s . However, both C_s and L_s introduce the imaginary part of the impedance. Therefore, L_g is added to compensate for the imaginary part. The value of L_s is very crucial as NF can be reduced by choosing a small L_s , while a smaller L_s also results in a larger voltage gain, degrading the linearity. By varying the value of L_s and making trade-offs between gain, NF, and also the linearity, the final setting is chosen to be $L_s = 900\text{pH}$, $C_s = 142\text{fF}$, and $L_g = 5.5\text{nH}$:

$$Z_{in(Ex: high gain)} = \frac{1}{s \times 142f} + s(5.5n + 900p) + \frac{14.44m \times 900p}{142f} = -23.1s + 91 \Omega \quad (8)$$

The theoretical result is still not accurate compared to the simulation results.

The output matching is quite easy as it's mainly determined by NM3 and R_1 . Varying the value of R_1 to let $R_{out} = 50 \Omega$ under the condition that NM3 is saturated. Finally, we get $R_1 = 80 \Omega$.

After that, the gain switching between the two modes is realized by adding a MOSFET (NM2) to adjust the current flow through the LNA. For the low-gain mode, the voltage V_B is connected to the VDD, where NM2 must be saturated so that a part of the current in the main amplifier branch will flow into NM2, thus decreasing the g_m , as well as the overall gain of the LNA. With a higher width of NM2, more current will be lost with lower gain as the result. Apart from the extra resistance load introduced, NM2 has little effect on the high-gain mode as it will be closed with V_B connected to the ground. Finally, the width for NM2 is chosen to be $36\mu\text{m}$ to satisfy the requirements.

At last, simulate the whole structure to confirm all specifications are met. Many trade-offs should be made in this part. For instance, the value for L_s that relates to linearity, NF, and gain. Also, to satisfy other requirements, the center frequency may not be set to the ideal value of 5G .

IV. SPECIFICATION VERIFICATION

A. High-Gain Mode Characteristics

Plots of the simulated performance in the high-gain mode are shown in Figures 4, 5, 6, 7, and 8 below.

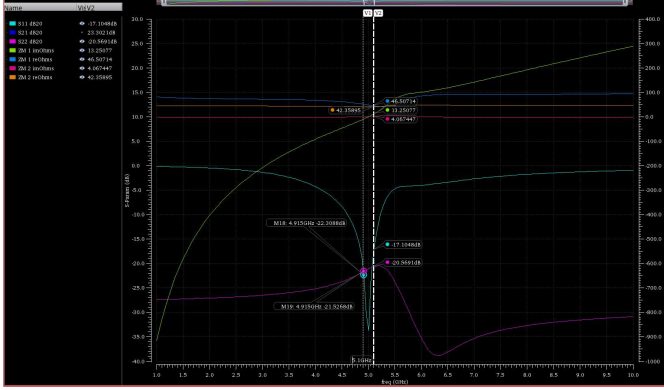


Fig. 4. S11 dB20 (blue) and S22 dB20 (purple) plot (S11 at 4.9GHz=-22.3088dB, S11 at 5.1GHz=-17.1048dB; S22 at 4.9GHz=-21.5268dB, S22 at 5.1GHz=-20.5691dB).

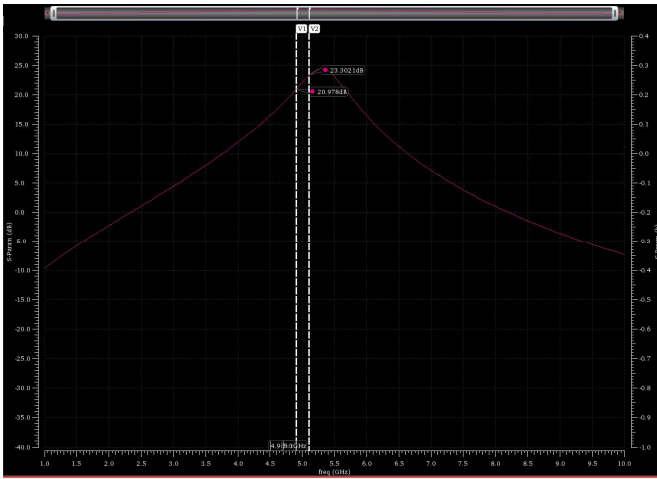


Fig. 5. S21 dB20 (power gain) plot (power gain at 4.9GHz=20.978dB; power gain at 5.1GHz=23.3021dB).

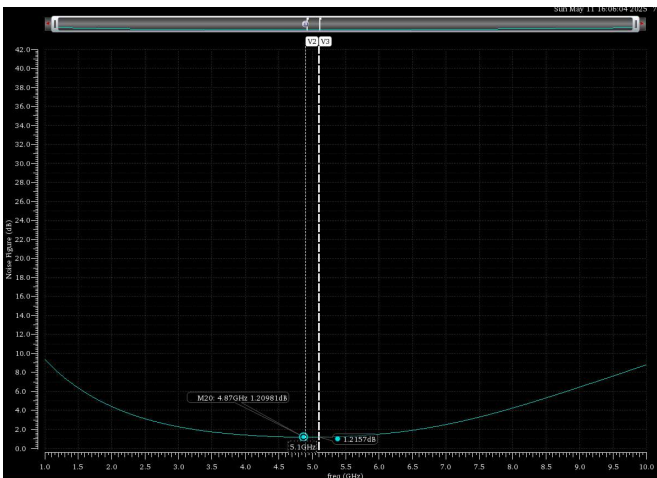


Fig. 6. Noise Figure plot (NF at 4.9GHz=1.2098dB; NF at 5.1GHz=1.2157dB).

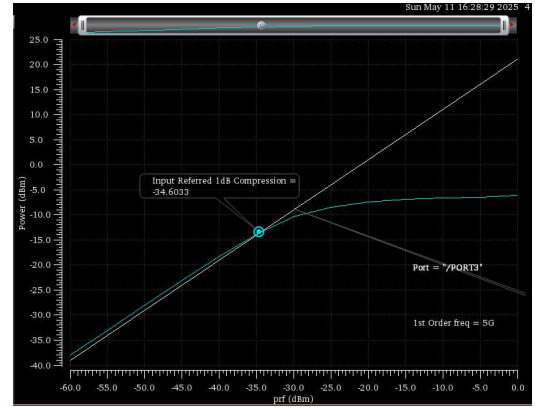


Fig. 7. P1dB plot (the input referred 1dB compression = -34.6033 dBm).

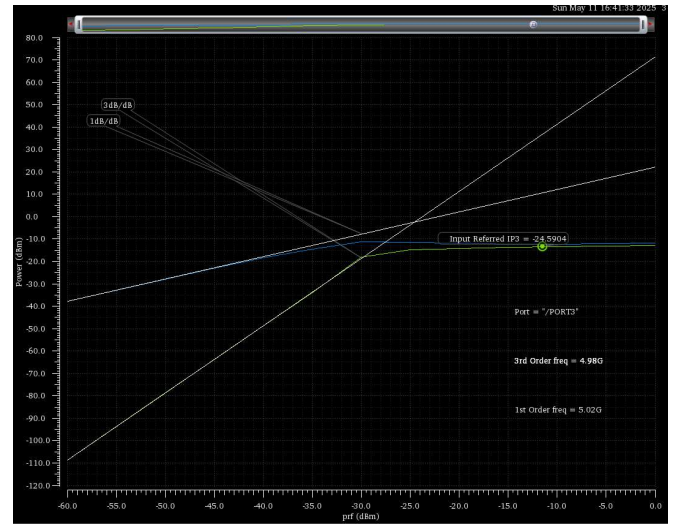


Fig. 8. IIP3 plot (the input referred IP3 = -24.5904 dBm).

Moreover, according to Figure 2, the power consumption can be calculated as:

$$VDD \times I = 1V \times 3.76822mA = 3.76822 \text{ mW} \quad (9)$$

B. Low-Gain Mode Characteristics

Plots of the simulated performance in the high-gain mode are shown in Figures 9, 10, 11, 12, and 13 below.

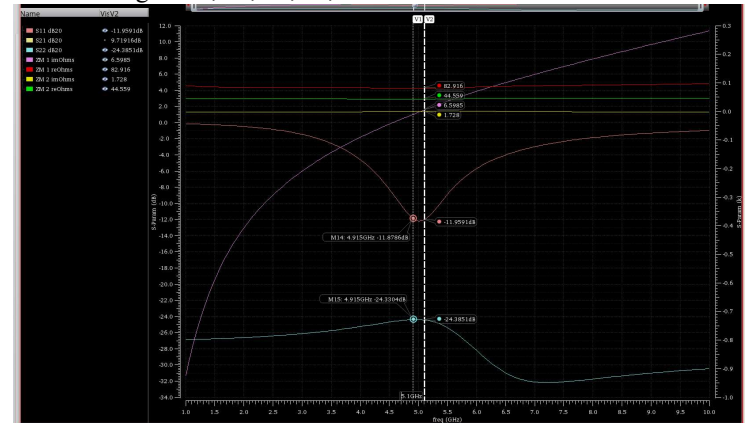


Fig. 9. S11 dB20 (orange) and S22 dB20 (blue) plot (S11 at 4.9GHz=-11.8786dB, S11 at 5.1GHz=-11.9591dB; S22 at 4.9GHz=-24.3304dB, S22 at 5.1GHz=-24.3851dB).

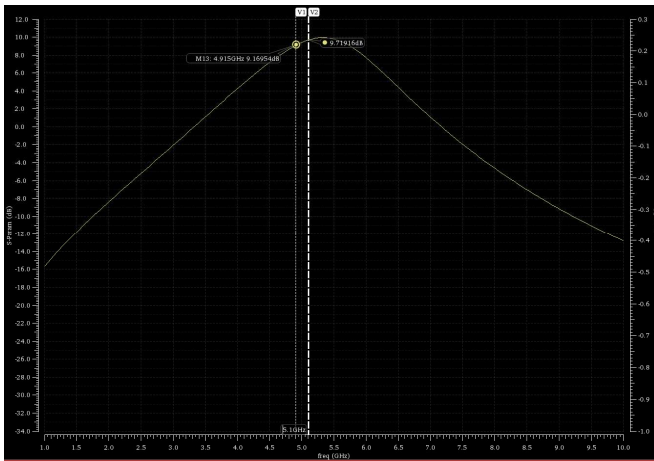


Fig. 10. S21 dB20 (power gain) plot (power gain at 4.9GHz=9.16954dB; power gain at 5.1GHz=9.71916dB).

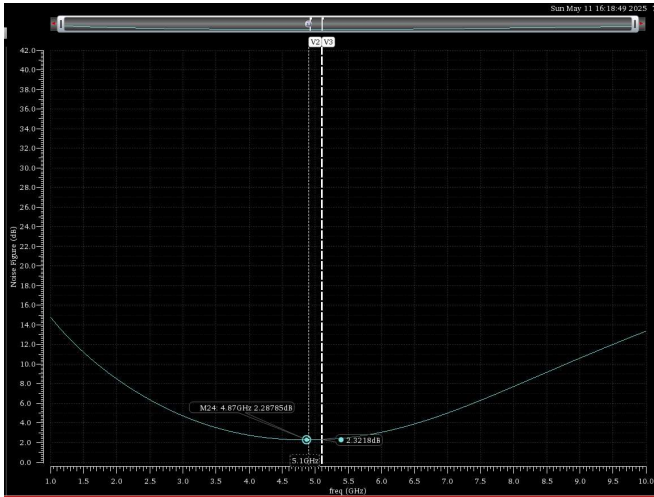


Fig. 11. Noise Figure plot (NF at 4.9GHz=2.28785dB; NF at 5.1GHz=2.3218dB).

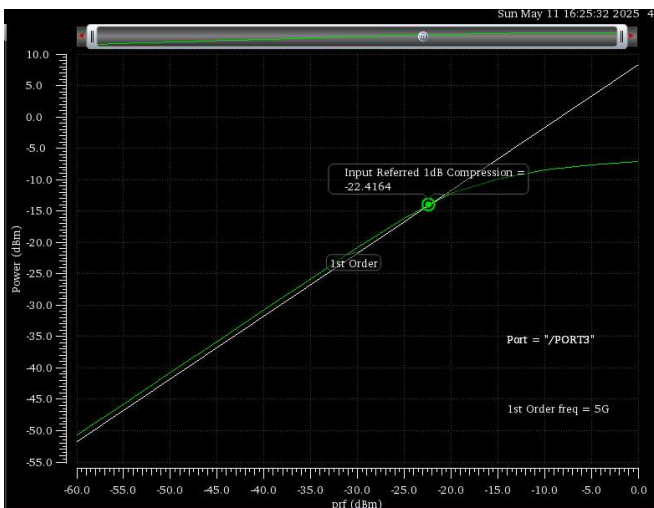


Fig. 12. P1dB plot (the input referred 1dB compression = -22.4164 dBm).

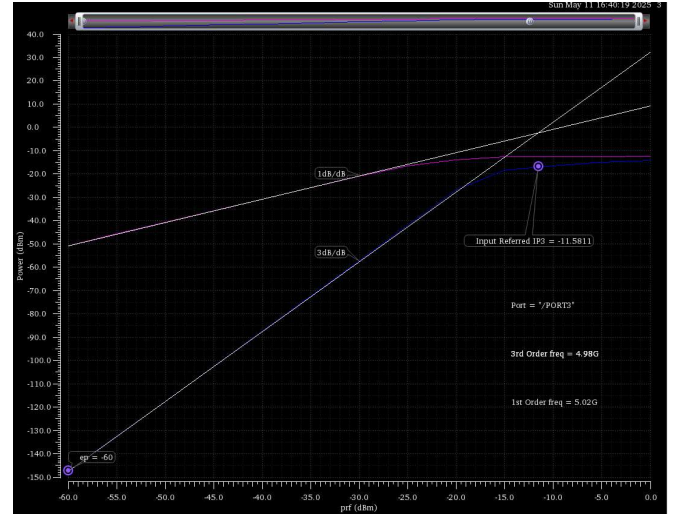


Fig. 13. IIP3 plot (the input referred IP3 = -11.5811 dBm).

Moreover, according to Figure 3, the power consumption can be calculated as:

$$VDD \times I = 1V \times 3.99546mA = 3.99546 \text{ mW} \quad (10)$$

C. Summary

In this case, the final results compared with the specifications are shown in Table 3.

TABLE III
COMPARISON BETWEEN SPECIFICATIONS AND SIMULATIONS

High-gain Mode			
Parameters	Specifications	Simulations	Calculations
Frequency Range	4.9 - 5.1 GHz	$f_0 = 5.32 \text{ GHz}$	$f_0 = 6.56 \text{ GHz}$
Input/Output Matching (S11 and S22)	$\leq -10\text{dB}$	S11 $\leq -17.1\text{dB}$ S22 $\leq -20.6\text{dB}$	
Power Gain (S21)	$\geq 20\text{dB}$	$\geq 21\text{dB}$	
NF	$\leq 1.5\text{dB}$	$\leq 1.215\text{dB}$	
P1dB	$\geq -36\text{dBm}$	-34.6033dBm	
IIP3	$\geq -26\text{dBm}$	-24.5904dBm	
Supply Voltage	1V		
Power Consumption	$\leq 3.8\text{mW}$	3.76822mW	
Low-gain Mode			
Parameters	Specifications	Simulations	Calculations
Frequency Range	4.9 - 5.1 GHz	$f_0 = 5.32 \text{ GHz}$	$f_0 = 6.56 \text{ GHz}$
Input/Output Matching (S11 and S22)	$\leq -10\text{dB}$	S11 $\leq -11.8\text{dB}$ S22 $\leq -24.3\text{dB}$	
Power Gain (S21)	$\leq 10\text{dB}$	$\leq 9.72\text{dB}$	
NF	$\leq 4\text{dB}$	$\leq 2.32\text{dB}$	
P1dB	$\geq -26\text{dBm}$	-22.4164dBm	
IIP3	$\geq -16\text{dBm}$	-11.5811dBm	
Supply Voltage	1V		
Power Consumption	$\leq 4.2\text{mW}$	3.99546mW	

V. FURTHER EXPLORATION

According to the requirement, change the signal frequency and simulate again the IIP3 by using two-tones at frequencies $f_{rf1}=4.5\text{GHz}$ and $f_{rf2}=5.5\text{GHz}$ (there may be some error for the description of the requirement, since the frequency of the input signal is 5GHz, the two-tone testing should be set on both sides of the center frequency). The result of the IIP3 plot is shown in Figure 14.

Compared to Figure 8, the observed IIP3 values of -24.44 dBm (4.5G/5.5G) and -24.59 dBm (4.98G/5.02G) show slight frequency-dependent nonlinearity. Wider 1 GHz spacing improves IIP3, likely due to filtering of unwanted signals by high-frequency effects. Narrower 40 MHz spacing worsens IIP3, indicating stronger memory effects within the amplifier. The small variation in IIP3 (<0.2 dB) suggests relatively consistent nonlinearity.

To get a better performance, it's important to improve the impedance and bias design to reduce memory effects at narrow spacing, and also manage high-frequency effects for wideband operation. This will help reduce unwanted signal distortion when signals are close together and maintain good performance across a wide range of frequencies. Overall, the circuit performs well, but it isn't perfectly linear in all situations.

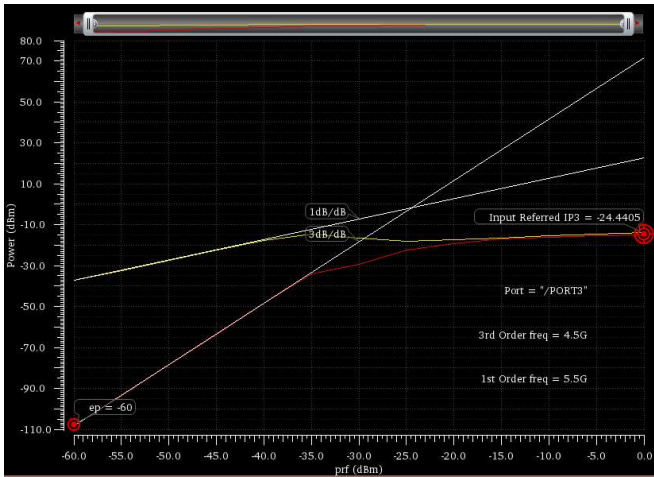


Fig. 14. IIP3 plot with $f_{rf1}=4.5\text{GHz}$ and $f_{rf2}=5.5\text{GHz}$ (the input referred IP3 = -24.4405 dBm).

New added: However, if the two tones are indeed located at $f_{rf1}=5.5\text{GHz}$ and $f_{rf2}=6\text{GHz}$, which are higher than the input signal frequency. The result of the IIP3 plot is shown in Figure 15, where the IIP3 drops steeply to -114.77dBm.

This is because when these two interference tones pass through, the LNA will introduce third-order intermodulation distortion (IM3) due to its nonlinearity. The IM3 frequencies are generated as:

$$f_{IM3} = 2f_{int1} - f_{in} = 11G - 6G = 5G \text{ Hz} \quad (11)$$

$$f_{IM3}' = 2f_{int2} - f_{int1} = 12G - 5.5G = 6.5G \text{ Hz} \quad (12)$$

In this case, IM3 frequencies at the LNA output are 5GHz and 6.5GHz, which overlap with the input signal frequency 5G, directly corrupting reception. This effectively reduces the receiver's effective IIP3.

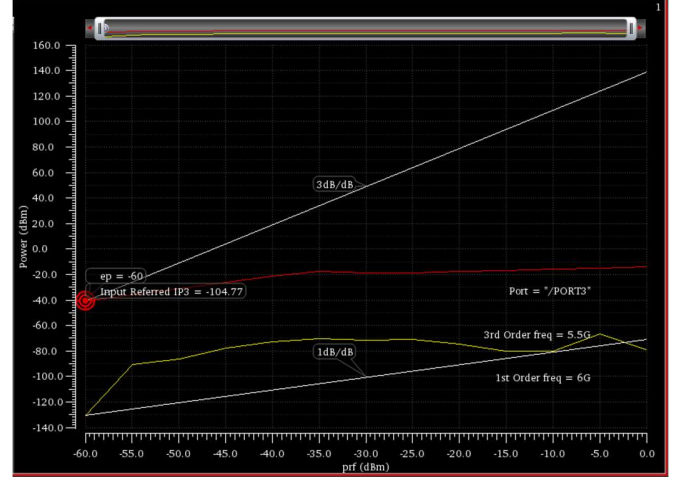


Fig. 15. IIP3 plot with $f_{rf1}=5.5\text{GHz}$ and $f_{rf2}=6\text{GHz}$ (the input referred IP3 = -104.77 dBm).

VI. CONCLUSION

This project successfully designed and simulated a low-noise amplifier (LNA) in Cadence, meeting the specified requirements. The report details both the theoretical calculations and simulation results, demonstrating the importance of careful parameter optimization to achieve a balanced and high-performing amplifier.

By comparing the theoretical and simulation results, it can also be found that there is a big difference between the actual results and the theoretical values. This shows that the theory only provides a rough guide, and the actual value determination still needs to be based on the theory with multiple simulations.

The key point of this project design is trade-offs. It's hard to satisfy the best performance of every characteristic. Therefore, trade-offs are needed to compensate each value and try to get the best overall performance.

In conclusion, this project has helped me to better understand the operation and structural features of the LNA. Moreover, it has laid the foundation for my future studies in analogue RF circuits

VII. ACKNOWLEDGEMENT

The author would like to thank Prof. Jun Yin, Kevin, and the TA, Mr. Xu, for the kind guidance and assistance to the author in this project.